
HIAL: Towards Semantics-Aware Hypergraph Active Learning via Dual-Perspective Information Maximization

Yanheng Hou^{*1} Xunkai Li^{*1} Yanzhe Wen¹ Zhenjun Li² Bing Zhou³ Rong-Hua Li¹ Guoren Wang¹

Abstract

Hypergraph Neural Networks (HNNs) model high-order interactions effectively but rely on costly node annotations, motivating Hypergraph Active Learning (HAL). However, many HAL pipelines adapt graph-based querying through clique expansion, which introduces *structural bias* and can cause *ranking collapse*, making utilities overly determined by hyperedge cardinalities rather than informative high-order context. We propose **HIAL** (Hypergraph Influence-based Active Learning), a training-free framework that formulates hypergraph active learning as influence maximization over a high-order context-based weighted pairwise projection of the hypergraph. HIAL employs a High-Order Interaction-aware propagation mechanism that modulates pairwise influence weights using hyperedge cardinality and feature consistency, yielding a scalable linear diffusion process tailored to homophilic hypergraphs. We further combine feature-space coverage and structural reachability into a monotone submodular selection objective, enabling an efficient lazy greedy solver. Experiments on eight benchmarks demonstrate that HIAL consistently outperforms strong baselines across diverse homophilic hypergraph domains.

1. Introduction

Hypergraphs provide a natural abstraction for modeling **high-order interactions (HOI)** in complex systems, such as protein complexes, group communications, and scientific collaborations. By explicitly encoding multi-way relations, Hypergraph Neural Networks (HNNs) often outperform

Graph Neural Networks (GNNs) that are restricted to pairwise edges. However, these advantages typically come with a practical bottleneck: obtaining high-quality node labels is expensive, time-consuming, and sometimes infeasible at scale. This motivates **Hypergraph Active Learning (HAL)**, which aims to select a small yet informative set of nodes to label so as to maximize downstream performance.

While Active Learning on simple graphs is established, extending these strategies to hypergraphs remains challenging. Early heuristic methods, such as AGE (Cai et al., 2017), combine centrality with uncertainty metrics, while partition-based strategies like GraphPart (Ma et al., 2023) utilize community structures. However, these approaches often rely on ad-hoc combinations of indicators or coarse partitions that may obscure individual node influence. Optimization-based methods, including GRAIN (Zhang et al., 2021b), RIM (Zhang et al., 2021a), and DOCTOR (Song et al., 2023), offer theoretical grounding but are designed for pairwise interactions. Applying these graph-based methods to hypergraphs typically necessitates Clique Expansion (CE). As illustrated in Figure 1, this projection introduces significant *Structural Bias*, leading to *Ranking Collapse*. By decomposing hyperedges into uniformly weighted cliques, CE creates a strong linear dependency between node influence and hyperedge size. This degree inflation overrides topological nuances, causing centrality metrics to prioritize nodes in large hyperedges regardless of their actual structural or semantic importance. Although hypergraph-specific AL methods exist, such as ACGNN (Sun et al., 2022), CGE-AL (Liao et al., 2023) and AHGA (Ni et al., 2026), they are predominantly model-dependent. These approaches rely on iterative model retraining to estimate node utility, which imposes computational overhead and limits their applicability to large-scale datasets. This highlights the need for a **training-free** framework capable of identifying informative nodes, independent of the downstream classifier’s training status.

To address these limitations, we introduce HIAL. We formulate the active node selection as an Influence Maximization (IM) proxy, where node selection aims to maximize the diffusion of label information. Unlike uniform CE, HIAL constructs a context-aware weighted pairwise projection.

^{*}Equal contribution ¹Beijing Institute of Technology, Beijing, China ²Shenzhen Institute of Technology, Shenzhen, China ³Shenzhen City Polytechnic, Shenzhen, China. Correspondence to: Rong-Hua Li <lironghuabit@126.com>.

By combining this propagation with a dual-perspective objective that considers both feature-space coverage and topological reachability, we derive a submodular function efficiently solvable via a lazy greedy strategy.

Our main contributions are summarized as follows:

(1) Semantics-Aware Weighted Projection. We propose a HAL framework based on a high-order-aware weighted pairwise projection. By jointly using inverse-cardinality normalization and within-hyperedge feature consistency, HIAL mitigates the ranking collapse issue observed in uniformly weighted CE methods.

(2) Dual-Perspective Objective. We design an objective function that selects nodes based on both structural influence and feature-space coverage, making the selected nodes representative and informative in homophilic hypergraphs.

(3) Scalability and Performance. We prove the submodularity of our objective, allowing for a theoretical $(1 - 1/e)$ approximation via a Lazy Greedy strategy. This enables HIAL to scale to large datasets with over 240K nodes while outperforming state-of-the-art baselines.

2. Preliminary

2.1. Hypergraphs and Clique Expansion

Consider a hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E})$, composed of a set of $n = |\mathcal{V}|$ nodes and a set of $m = |\mathcal{E}|$ hyperedges. Each hyperedge $e \in \mathcal{E}$ is a non-empty subset of nodes, allowing for the modeling of high-order correlations beyond pairwise connections. The structure is represented by an incidence matrix $\mathbf{H} \in \{0, 1\}^{n \times m}$, where $H_{ve} = 1$ if node $v \in e$, and 0 otherwise. Additionally, a diagonal matrix $\mathbf{W} \in \mathbb{R}^{m \times m}$ assigns a weight w_e to each hyperedge.

A common approach to processing hypergraphs is Clique Expansion (CE), which decomposes each hyperedge into a clique of pairwise edges. The adjacency matrix of the induced graph, denoted as $\mathbf{A}_{CE}$, is formally defined as:

$$\mathbf{A}_{CE} = \mathbf{H}\mathbf{W}\mathbf{H}^\top - \text{diag}(\mathbf{H}\mathbf{W}\mathbf{H}^\top) \quad (1)$$

where the diagonal matrix is subtracted to eliminate self-loops. While uniform CE enables standard graph algorithms, it may flatten structure. This transformation risks making pairwise utilities overly determined by hyperedge cardinality, a phenomenon we characterize as *Ranking Collapse* (analyzed in Sec. 3.1).

2.2. Hypergraph Active Learning (HAL)

Given an unlabeled pool $\mathcal{V}_{train}$ and a labeling budget $\mathcal{B}$, the goal of HAL is to select a subset $\mathcal{S} \subset \mathcal{V}_{train}$ ($|\mathcal{S}| = \mathcal{B}$) that minimizes the generalization error of a Hypergraph Neural Network (HNN) trained on $\mathcal{S}$.

$$\min_{\mathcal{S} \subset \mathcal{V}_{train}, |\mathcal{S}| = \mathcal{B}} \mathbb{E}_{v_i \in \mathcal{V}_{test}} [\ell(y_i, P(\hat{y}_i | x_i; \mathcal{M}_{\mathcal{S}}))] \quad (2)$$

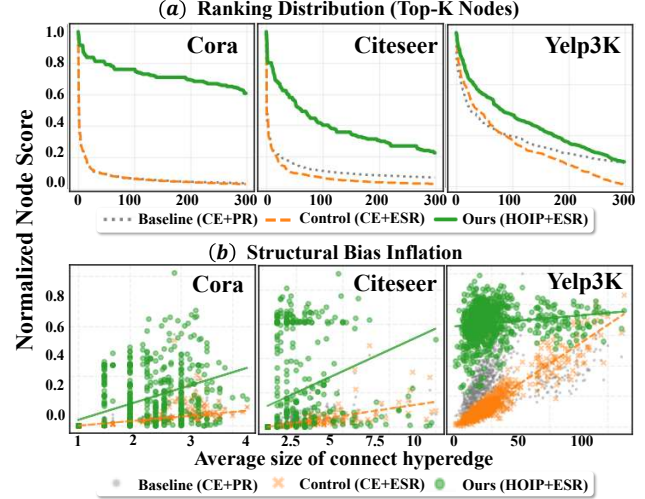


Figure 1. Analysis of Ranking Collapse and Structural Bias. **(a) Ranking Distribution.** Baseline and Control methods exhibit a sharp ranking collapse, failing to distinguish the majority of nodes. In contrast, HOIP produces a graded distribution, preserving ranking resolution. **(b) Structural Bias.** CE-based methods demonstrate a strong linear dependency on hyperedge size, indicating structural determinism. HOIP shows significant vertical dispersion, reflecting its ability to differentiate nodes based on semantic consistency rather than structural cardinality.

Current HAL research primarily follows two paradigms. Model-dependent methods (e.g., ACGNN) rely on computationally prohibitive iterative HNN training to query instances. Theoretical approaches like HS² (Chien et al., 2019) focus on graph cuts using pairwise queries, which diverges from the standard pointwise annotation setting required for training HNNs. In contrast, our work introduces a *training-free* paradigm by formalizing data selection as an influence maximization proxy, avoiding costly retraining loops while exploiting semantically high-order context.

2.3. Social Influence Maximization (SIM)

The SIM problem, first formalized by Kempe et al. (Kempe et al., 2003), seeks a seed set $\mathcal{S}$ that maximizes the spread of influence $\sigma(\mathcal{S})$ under a diffusion model:

$$\max_{\mathcal{S} \subset \mathcal{V}, |\mathcal{S}| = \mathcal{B}} \sigma(\mathcal{S}) \quad (3)$$

Connection to Active Learning. A useful analogy exists between GAL and SIM: selected nodes act as seeds, and label propagation can be viewed as an influence diffusion process (Xie et al., 2022; Zhang et al., 2021b;a; Guillory & Bilmes, 2011). We extend this analogy to hypergraphs and use influence maximization as a training-free proxy for improving data distribution coverage.

While the optimization is generally NP-hard, if $\sigma(\cdot)$ is monotone and submodular, a greedy strategy guarantees a $(1 - 1/e)$ -approximation (Nemhauser et al., 1978). Crucially,

this property enables the **Lazy Greedy** algorithm (Minoux, 2005), which avoids re-evaluating all marginal gains. This efficiency is pivotal for HIAL, as we will demonstrate that our proposed surrogate objective satisfies these conditions.

3. The Proposed HIAL Framework

In this section, we present the details of the HIAL framework, illustrated in Figure 3. Given a hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E})$ and a node feature matrix $\mathbf{X}$, our method proceeds in two main stages. (1) Constructing a **High-Order Interaction (HOI) Aware Propagation** mechanism that builds a weighted projection; (2) Optimizing a **Dual-Perspective Influence** objective that fuses structural reachability (ESR) with feature-space coverage (MFS) to select the most informative nodes without structural artifacts.

3.1. HOI-Aware Propagation Mechanism

The effectiveness of Hypergraph Neural Networks (HNNs) primarily stems from their ability to aggregate information across the hypergraph structure. Recent studies suggest that this power is derived mainly from feature smoothing over hypergraph neighborhoods rather than complex non-linear transformations (Rossi et al., 2020; He et al., 2020; Kim et al., 2024; Tang et al., 2025). Generalizing this insight, we establish a parameter-free linear diffusion mechanism to serve as the foundation for quantifying influence.

Standard Decoupled HNN Propagation. A standard HNN propagation layer typically employs a transition matrix $\mathbf{T}$ derived solely from hypergraph structure (e.g., $\mathbf{T} = \mathbf{D}_v^{-1/2} \mathbf{H} \mathbf{W} \mathbf{D}_e^{-1} \mathbf{H}^\top \mathbf{D}_v^{-1/2}$). Such propagation often treats interactions uniformly or relies on simple normalization, which can induce CE-style pairwise effects that are insensitive to within-hyperedge semantic consistency. Such methods suffer from *Structural Determinism*, where influence scores are dictated by connectivity quantity rather than interaction quality.

To analyze the mechanism of ranking collapse, we examined the correlation between average connected hyperedge size and influence scores, as shown in Figure 1. We evaluated three configurations: (1) Baseline (CE+PR), representing standard PageRank on Clique Expansion; (2) Control (CE+ESR), applying our ESR scoring function on the uniformly clique-expanded structure; and (3) Ours (HOIP+ESR), applying ESR on the HOIP projection.

As shown in Figure 1a, both Baseline and Control methods exhibit a rapid score decay, assigning near-zero scores to the majority of nodes. Figure 1b reveals the underlying cause: these methods show a strong linear dependency between influence scores and hyperedge size with minimal variance. This indicates that the scoring is dominated by structural cardinality, neglecting semantic context.

Table 1. Comparison of Structural Bias Coefficient (SBC). SBC measures the rank correlation between influence scores and the average cardinality of incident hyperedges. HOIP yields lower SBC values, indicating reduced dependence on hyperedge-size bias.

Dataset	Cora	Citeseer	Yelp3K
Baseline (CE+PR)	0.9785	0.9561	0.8311
Control (CE+ESR)	0.9988	0.9992	0.9993
Ours (HOIP+ESR)	0.8856	0.8936	0.6087

In contrast, HOIP produces a more graded distribution (Figure 1a) and displays significant vertical dispersion at fixed hyperedge sizes (Figure 1b). This variance suggests that HOIP differentiates nodes based on feature consistency rather than hyperedge size alone. We quantify this effect using the Structural Bias Coefficient (SBC), defined as the Spearman rank correlation between the computed influence scores and the average cardinality of incident hyperedges:

$$\text{SBC} = \rho(\text{rank}(\mathbf{s}), \text{rank}(\mathbf{h})) \quad (4)$$

where $\mathbf{s} \in \mathbb{R}^n$ is the vector of computed influence scores and $\mathbf{h} \in \mathbb{R}^n$ denotes the average cardinality of hyperedges incident to each node.

A high SBC indicates that the algorithm is merely replicating topological centrality, while a lower SBC suggests a successful decoupling of influence from raw connectivity. As presented in Table 1, baseline methods exhibit high SBC values, confirming a strong structural dependence. HOIP consistently reduces the SBC, indicating a successful decoupling of influence estimation from raw connectivity.

HOI-aware Propagation (HOIP). To address this limitation, we design a propagation mechanism that calibrates the intensity of high-order interactions. Instead of treating all connections uniformly, we quantify the interaction by integrating inverse-cardinality normalization with semantic consistency weighting. This approach decouples the propagation strength from the structural scale of the hyperedge.

Definition 3.1. (Interaction Intensity) The interaction intensity between node i and j is defined as:

$$l_{ij} = \sum_{e \in \mathcal{E}: d(e) > 1} w_e H_{ie} H_{je} \cdot \underbrace{\frac{1}{d(e) - 1}}_{\text{Structural}} \cdot \underbrace{\exp(-\sigma_e^2)}_{\text{Semantic}}, \quad i \neq j \quad (5)$$

where w_e is the weight of hyperedge e , $d(e) = \sum_{v \in \mathcal{V}} H_{ve}$ denotes the degree of hyperedge e . In implementation, σ_e^2 is computed from normalized features within hyperedge e .

Equation (5) introduces two terms to calibrate the interaction. First, the term $(d(e) - 1)^{-1}$ acts as a structural normalizer to prevent score inflation in large hyperedges. Second, and crucially, the term $\exp(-\sigma_e^2)$ serves as a

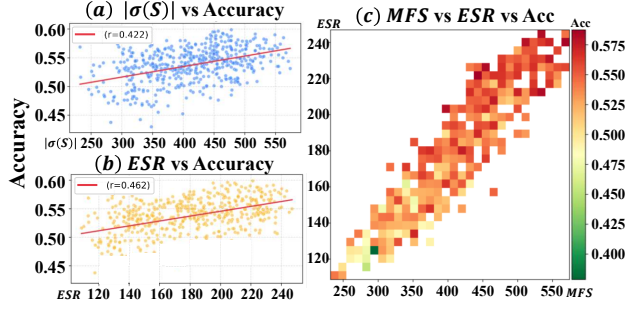


Figure 2. Impact of seed set quality on downstream performance (Cora, $|S| = 20C$). Plots (a) and (b) demonstrate positive correlations between our proposed influence metrics and model accuracy. Plot (c) reveals that maximizing both feature sensitivity (MFS) and structural reachability (ESR) yields higher accuracy.

semantic filter. Unlike topology reconstruction tasks that aim to preserve all connections (including heterophilic ones), the goal of active learning here is node classification via label propagation. In this context, propagating labels across high-variance hyperedges introduces noise and degrades decision boundaries. Therefore, HIAL adopts an explicit **homophily prior**: it attenuates structurally present but semantically divergent connections and focuses diffusion on channels where feature smoothness is expected to support node classification. This design scopes HIAL to homophilic hypergraphs. In strongly heterophilic settings, high-variance hyperedges may encode essential cross-class interactions, and the semantic attenuation term can become harmful. We discuss this limitation in Appendix H.

We then apply symmetric normalization to obtain $\tilde{\mathbf{L}} = \mathbf{D}^{-1/2} \mathbf{L} \mathbf{D}^{-1/2}$, where $\mathbf{D}$ is the diagonal degree matrix with entries $d_{ii} = \sum_j l_{ij}$. The final propagation is defined as:

$$\mathbf{X}^{(t+1)} = \mathbf{P} \mathbf{X}^{(t)} + (1 - \alpha) \mathbf{X}^{(0)}, \quad \mathbf{P} = \alpha \tilde{\mathbf{L}}. \quad (6)$$

where α controls the strength of propagation strength.

After k propagation steps, Eq. (6) induces a linear mapping from the initial features to the propagated features:

$$\mathbf{X}^{(k)} = \mathbf{A}_k \mathbf{X}^{(0)}, \quad \mathbf{A}_k = \mathbf{P}^k + (1 - \alpha) \sum_{t=0}^{k-1} \mathbf{P}^t. \quad (7)$$

We refer to $\mathbf{A}_k$ as the effective k -step diffusion operator. For the default setting $k = 1$, this reduces to $\mathbf{A}_1 = \mathbf{P} + (1 - \alpha) \mathbf{I}$.

Remark. HOIP deliberately uses a weighted pairwise diffusion form for scalability, but its weights are not produced by uniform clique expansion. Instead, each pairwise interaction is calibrated by hyperedge-level cardinality and feature consistency, allowing the diffusion process to retain informative high-order context while avoiding structural inflation from large hyperedges. The inverse-cardinality term suppresses size-induced degree bias, whereas the

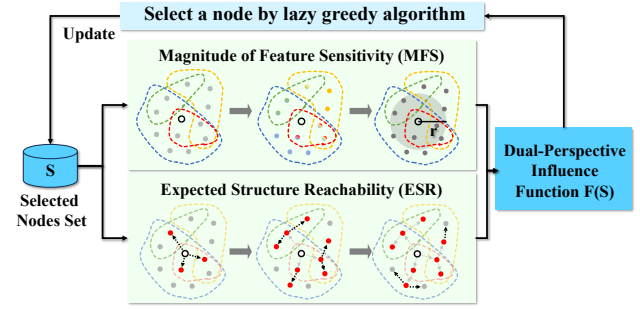


Figure 3. The workflow of HIAL. The framework combines Feature Sensitivity (MFS) and Structure Reachability (ESR) to quantify node influence, employing a lazy greedy algorithm for efficient seed set selection.

variance-based term emphasizes feature-consistent channels that are more reliable for homophilic node classification.

3.2. Dual-Perspective Influence Function

We now propose a dual-perspective surrogate function to quantify node influence from two aspects: (1) **Feature Sensitivity**, which measures semantic representativeness in propagated space, and (2) **Structural Reachability**, which measures diffusion capacity over neighborhoods.

1. Feature Sensitivity Inspired by sensitivity-based influence measures (Wang & Leskovec, 2020; Xu et al., 2018), we quantify how strongly the propagated representation of a target node depends on the initial feature of a candidate seed node under HOIP. Since HOIP is a linear diffusion process, this sensitivity can be computed directly from the diffusion operator, without model training or back-propagation.

Definition 3.2. (Feature Sensitivity) Let $\mathbf{A}_k$ be the effective k -step diffusion operator in Eq. (7). We define the influence from node i to node j after k steps as:

$$I(j, i, k) = |[\mathbf{A}_k]_{ji}| \quad (8)$$

This score measures the scalar node-to-node sensitivity induced by the linear HOIP diffusion.¹ To ensure comparability across target nodes with different total incoming influence, we adopt a normalized influence score:

$$I_j(i, k) = \frac{I(j, i, k)}{\sum_{v \in \mathcal{V}} I(j, v, k)} \quad (9)$$

For a seed set S , we adopt a **maximum-impact assumption**: a node is considered “activated” only if the strongest signal it receives from S exceeds a threshold θ .

¹Equivalently, for node features of dimension F , the block Jacobian satisfies $\partial \mathbf{X}_j^{(k)} / \partial \mathbf{X}_i^{(0)} = [\mathbf{A}_k]_{ji} \mathbf{I}_F$. Thus, the entrywise L_1 norm of this block Jacobian is proportional to $|[\mathbf{A}_k]_{ji}|$, and the constant factor is canceled by the normalization in Eq. (9).

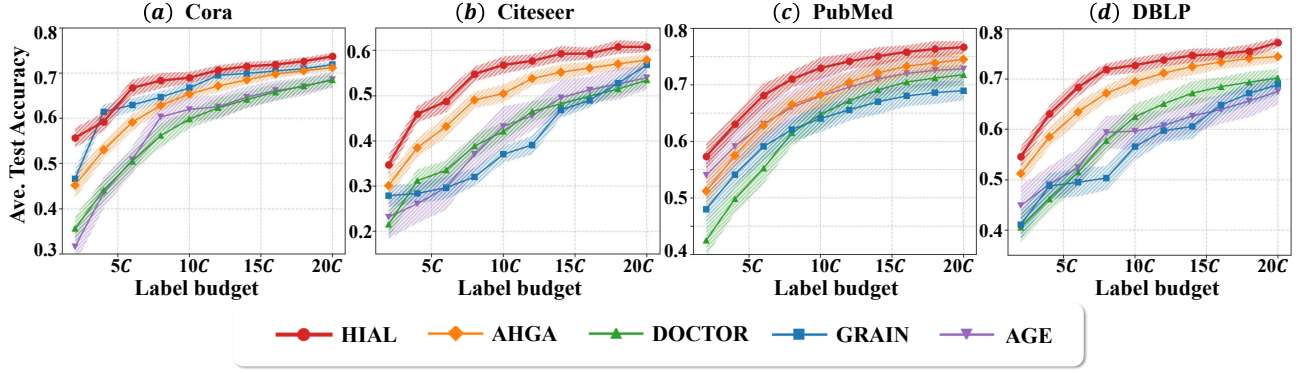


Figure 4. Test accuracy comparison on four datasets with labeling budget $\mathcal{B}$ ranging from $2C$ to $20C$. HIAL achieves the best performance with lower variance (shaded area) compared to baselines.

Definition 3.3. (Activated Set) Given a threshold θ , the set-to-point influence is defined as $I_j(S, k) = \max_{u \in S} I_j(u, k)$. The set of nodes **activated** by seed set S is:

$$\sigma(S) = \{j \in \mathcal{V} \mid I_j(S, k) > \theta\} \quad (10)$$

From Count to Spatial Coverage. A natural heuristic is that activating more nodes should correlate with better downstream performance. To verify this, we conducted analysis (Figure 2) by training HNNs with randomly selected seeds. While we observe a general positive correlation between $|\sigma(S)|$ and accuracy, Figure 2(a) reveals a crucial subtlety: seed sets with the same activation scale can lead to different accuracy. This implies that simply counting nodes ignores information redundancy; activating a cluster of similar nodes yields diminishing returns. To capture this, we model the influence of an activated node u as covering a ball G_u of radius r in the feature space:

$$G_u = \left\{v \in \mathcal{V} \mid d\left(X_u^{(k)}, X_v^{(k)}\right) \leq r\right\} \quad (11)$$

where $d(\cdot, \cdot)$ is the cosine distance.

Definition 3.4. Magnitude of Feature Sensitivity (MFS) of a seed set S is defined as the number of nodes in the union of covered regions:

$$MFS(S) = \left| \bigcup_{u \in \sigma(S)} G_u \right| \quad (12)$$

Remark. This formulation effectively penalizes redundancy: if multiple activated nodes cover the same feature region, the marginal gain in MFS is minimal. Two structurally distant but feature-similar nodes may cover overlapping regions in MFS , because querying both can provide redundant semantic information under a scarce labeling budget. When $r = 0$, $MFS(S)$ reduces to counting distinct activated feature points; if no two propagated node features are identical, this is equivalent to $|\sigma(S)|$.

Theorem 3.5. $MFS(S)$ is a monotone non-decreasing and submodular function. (Proof in Appendix A).

2. Structural Reachability While MFS captures semantic coverage in the propagated feature space, it does not explicitly account for local neighborhood activation. To complement it, we model how information diffuses through hyperedge neighborhoods. As shown in Figure 1, measures based on expansions suffer from Structural Distortion. Therefore, we propose Expected Structure Reachability, which models the *probabilistic activation* of neighbors.

Definition 3.6. Expected Structure Reachability (ESR). Let $\mathcal{N}(v)$ denote the one-hop neighbor set of node v (excluding itself), and define $\mathcal{N}(S) = (\bigcup_{u \in S} \mathcal{N}(u)) \setminus S$. The ESR of a seed set S is defined as

$$ESR(S) = |S| + \sum_{v \in \mathcal{N}(S)} \left[1 - \prod_{u \in S \cap \mathcal{N}(v)} (1 - p_{uv}) \right] \quad (13)$$

where the term $1 - \prod_{u \in S \cap \mathcal{N}(v)} (1 - p_{uv})$ follows a noisy-or model, i.e., the probability that v is activated by *at least one* neighbor in S under an (approximate) conditional independence assumption.

We set the transmission probability as $p_{uv} = \beta \cdot \frac{c(u, v)}{d(u)}$ where $c(u, v)$ is the co-occurrence count between u and v , and $d(u) = \sum_{v \in \mathcal{N}(u)} c(u, v)$. With $\beta \in (0, 1]$, this construction ensures $p_{uv} \in [0, 1]$. We use $\beta = 0.5$ as a neutral prior to model information decay in the absence of learned diffusion weights.

Remark. Eq. (13) models collective activation induced by hyperedge co-membership. The formulation captures the diminishing returns of adding structurally redundant seeds.

Theorem 3.7. $ESR(S)$ is non-decreasing and submodular. (Proof in Appendix A)

3.3. Unified Selection Algorithm

We combine feature sensitivity (MFS) and structure reachability (ESR) into a unified objective. The two terms are related but not redundant. MFS measures sensitivity-triggered coverage in the propagated feature space, penalizing semantically redundant activations even when nodes are structurally distant. ESR , in contrast, measures local structural reachability induced by hyperedge co-membership and penalizes overlapping neighborhoods. Thus, MFS acts as a semantic redundancy control, while ESR preserves local diffusion capacity. We use their combination as a training-free surrogate for label efficiency rather than a mechanistic guarantee of downstream accuracy.

Although both metrics quantify the influence in terms of node counts, they operate at vastly different scales due to the density mismatch between the smoothed feature space and the sparse graph structure. A single node often covers a large cluster in the feature space (high MFS) but connects to fewer neighbors structurally (low ESR). To resolve this and ensure balanced contribution, we employ Initial Scale Normalization:

$$F(S) = \gamma \cdot \frac{MFS(S)}{\Psi_{MFS}} + (1 - \gamma) \cdot \frac{ESR(S)}{\Psi_{ESR}} \quad (14)$$

where $\Psi_{MFS} = \max_{v \in \mathcal{V}} MFS(\{v\})$ and $\Psi_{ESR} = \max_{v \in \mathcal{V}} ESR(\{v\})$ are the maximum marginal gains observed in the first iteration. We set $\gamma = 0.5$ as default. Since Eq. (14) is a positive linear combination of two monotone submodular functions, the unified objective $F(S)$ remains monotone submodular. We thus employ the Lazy Greedy optimization (Minoux, 2005) (Algorithm 1) to maximize $F(S)$. Since the scaling factors Ψ are fixed after initialization, marginal gains remain comparable across iterations. This allows Lazy Greedy pruning to be applied directly. The algorithm therefore inherits the $(1 - 1/e)$ approximation guarantee for the surrogate objective.

4. Experiments

To validate the effectiveness and efficiency of the HIAL framework for hypergraph data selection, we design experiments to address the following key questions:

Q1: Effectiveness. Does HIAL outperform state-of-the-art baselines in hypergraph active learning (HAL)?

Q2: Interpretability. Do the core components of HIAL (MFS , ESR , and HOI propagation) contribute to its performance?

Q3: Efficiency and Scalability. Is HIAL efficient and scalable for large-scale hypergraphs?

Q4: Generalization. Can HIAL perform well with different HNN models?

Q5: Robustness. How sensitive is HIAL to hyperparameters?

Algorithm 1 HIAL Node Selection Strategy

Require: Hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E})$, Features $\mathbf{X}$, Budget $\mathcal{B}$, Steps k , Balance γ .

Ensure: Selected Seed Set S .

```

1: // Stage 1: HOI-Aware Structural Preparation
2: Compute HOIP interaction weights via Eq. (5);
3: Prepare sparse normalized propagation access  $\tilde{\mathbf{L}}$ ;
4: for  $i = 1$  to  $k$  do
5:    $\mathbf{X}^{(i)} = \alpha \tilde{\mathbf{L}} \mathbf{X}^{(i-1)} + (1 - \alpha) \mathbf{X}^{(0)}$ ;
6: end for
7: Prepare column-wise influence evaluation and normalization factors for Eq. (9);
8: // Stage 2: Normalization Setup
9: Compute static scaling factors:
10:  $\Psi_{MFS} \leftarrow \max_{v \in \mathcal{V}} MFS(\{v\})$ 
11:  $\Psi_{ESR} \leftarrow \max_{v \in \mathcal{V}} ESR(\{v\})$ ;
12: // Stage 3: Dual-Perspective Selection
13: Initialize  $S = \emptyset$ ;
14: while  $|S| < \mathcal{B}$  do
15:   Define Marginal Gain for any candidate  $v$ :
16:    $\Delta_{MFS}(v) = MFS(S \cup \{v\}) - MFS(S)$ ;
17:    $\Delta_{ESR}(v) = ESR(S \cup \{v\}) - ESR(S)$ ;
18:    $\Delta(v) = \gamma \frac{\Delta_{MFS}(v)}{\Psi_{MFS}} + (1 - \gamma) \frac{\Delta_{ESR}(v)}{\Psi_{ESR}}$ ;
19:   Select  $v^*$  using Lazy Greedy marginal-gain updates;
20:    $S \leftarrow S \cup \{v^*\}$ ;
21: end while
22: return  $S$ 

```

4.1. Settings

Datasets. To comprehensively evaluate the effectiveness and scalability of HIAL, we conduct experiments on **eight** real-world hypergraph datasets, covering diverse domains and scales. These include three citation networks (**Cora**, **Citeseer**, **PubMed**), one co-authorship network (**DBLP**), two social/text networks (**Yelp3K**, **News20**), one biological dataset (**Mushroom**), and one large-scale culinary network (**Recipe200K**). More details are provided in the Appendix B.

Baselines. We compare HIAL against a comprehensive set of state-of-the-art baselines, categorized into three groups: (1) *Heuristic Methods*: **Random**, **Degree-N** (Node Degree), and **Degree-HE** (Hyperedge Degree); (2) *Learning-based Methods*: **AGE** (Cai et al., 2017), **CGE-AL** (Liao et al., 2023), **ACGNN** (Sun et al., 2022), and **AHGA** (Ni et al., 2026), which rely on model feedback or iterative retraining; (3) *Influence & Structure-based Methods*: **GraphPart** (Ma et al., 2023), **DOCTOR** (Song et al., 2023), **GRAIN** (Zhang et al., 2021b), and **RIM** (Zhang et al., 2021a), which focus on topological coverage or influence maximization. More detailed information is provided in the Appendix C.

Implementation Details. For the downstream classification

Table 2. Node classification accuracy (Mean $\pm$ Std, %) of active learning methods on HNNs. The labeling budget is set to $B = 20C$ for small-scale datasets and $50C$ for large-scale datasets, where C is the number of classes. **Bold** and underlined values indicate the best and second-best results, respectively. “OOT” and “OOM” denote Out-of-Time and Out-of-Memory exceptions.

Method	Cora	Citeseer	DBLP	PubMed	Yelp3K	News20	Mushroom	Recipe200K
Random	65.27 $\pm$ 2.30	45.77 $\pm$ 2.50	61.97 $\pm$ 1.70	47.77 $\pm$ 1.30	61.62 $\pm$ 1.25	62.15 $\pm$ 1.85	55.00 $\pm$ 1.80	56.07 $\pm$ 1.10
Degree-N	59.78 $\pm$ 1.40	51.32 $\pm$ 1.80	66.68 $\pm$ 1.15	70.01 $\pm$ 2.90	69.45 $\pm$ 0.85	65.87 $\pm$ 1.42	73.25 $\pm$ 1.20	54.26 $\pm$ 0.95
Degree-HE	66.91 $\pm$ 0.95	46.73 $\pm$ 1.00	65.71 $\pm$ 1.65	61.82 $\pm$ 4.50	64.07 $\pm$ 0.92	64.95 $\pm$ 1.50	70.87 $\pm$ 1.35	53.24 $\pm$ 0.98
AHGA	71.21 $\pm$ 0.75	<u>57.93$\pm$1.30</u>	<u>74.55$\pm$1.20</u>	<u>75.21$\pm$1.50</u>	95.81 $\pm$ 0.45	74.22 $\pm$ 0.95	88.45 $\pm$ 0.65	OOM
CGE-AL	62.73 $\pm$ 1.30	48.51 $\pm$ 1.10	64.51 $\pm$ 2.68	66.76 $\pm$ 4.80	98.12 $\pm$ 0.38	76.55 $\pm$ 1.10	77.86 $\pm$ 0.72	55.60 $\pm$ 1.25
ACGNN	67.81 $\pm$ 1.20	53.87 $\pm$ 1.40	67.83 $\pm$ 1.80	73.20 $\pm$ 1.55	96.87 $\pm$ 0.42	78.10 $\pm$ 1.05	90.88 $\pm$ 0.55	56.93 $\pm$ 1.30
DOCTOR	68.45 $\pm$ 1.50	53.45 $\pm$ 1.50	70.20 $\pm$ 1.80	71.82 $\pm$ 1.80	98.45 $\pm$ 0.35	81.25 $\pm$ 0.88	94.17 $\pm$ 0.48	57.15 $\pm$ 1.15
GraphPart	65.53 $\pm$ 1.10	49.14 $\pm$ 1.44	66.58 $\pm$ 1.05	69.12 $\pm$ 1.70	97.80 $\pm$ 0.40	79.62 $\pm$ 0.90	92.54 $\pm$ 0.52	56.41 $\pm$ 1.08
GRAIN	<u>71.90$\pm$0.70</u>	56.81 $\pm$ 1.30	68.92 $\pm$ 2.00	68.97 $\pm$ 1.70	<u>98.90$\pm$0.25</u>	<u>82.40$\pm$0.75</u>	94.51 $\pm$ 0.35	<u>57.86$\pm$0.85</u>
RIM	68.99 $\pm$ 1.00	57.54 $\pm$ 1.10	68.68 $\pm$ 2.70	70.13 $\pm$ 0.50	96.25 $\pm$ 0.30	OOT	95.50 $\pm$ 0.40	OOT
AGE	68.64 $\pm$ 1.80	53.98 $\pm$ 2.60	67.53 $\pm$ 2.60	72.82 $\pm$ 2.20	98.88 $\pm$ 0.55	OOT	63.44 $\pm$ 0.85	OOT
HIAL	72.97$\pm$0.68	61.55$\pm$1.35	78.41$\pm$0.84	76.14$\pm$1.37	99.21$\pm$0.42	87.66$\pm$1.65	97.29$\pm$1.17	60.21$\pm$1.95

task, we employ a standard decoupled HGNN as the backbone model for all methods to ensure a fair comparison. The labeling budget B is set to $20C$ (where C is the number of classes) for small-scale datasets and $50C$ for the large-scale dataset. We report the mean test accuracy and standard deviation over 10 independent runs with different random seeds. All experiments are implemented in PyTorch and executed on a server with an Intel Xeon Gold CPU and an NVIDIA RTX 4090 GPU (24GB).

Code Availability. The source code and implementations are published in <https://github.com/houhyh/icml26-HIAL>.

Efficiency Metrics. OOM denotes exceeding the 24GB GPU memory limit of the NVIDIA RTX 4090. OOT denotes exceeding a 24-hour wall-clock time limit. Runtime measures the active selection phase only, excluding downstream HNN training after the labeled set is fixed. For model-dependent baselines, the selection phase includes the iterative retraining required to update acquisition scores, because such retraining is part of their acquisition procedure.

Hyperparameter Settings. For all baselines, we utilize the optimal hyperparameters reported in their original papers or fine-tune them using the provided official codebases to achieve their best performance. Concrete settings for HIAL are shown in Appendix E.

4.2. Performance Comparison (Q1)

We evaluated HIAL against baselines by tracking test accuracy across increasing labeling budgets (Figure 4). HIAL exhibits a significantly steeper learning curve, indicating superior data efficiency.

Superior Accuracy. As detailed in Table 2, HIAL consistently achieves state-of-the-art performance. On Citeseer, HIAL surpasses AHGA by 3.62% and RIM by

4.01%. On Cora, Yelp3K and News20, HIAL outperforms competitive baseline GRAIN. Notably, on the large-scale Recipe200K dataset, HIAL outperforms GRAIN by 2.35%, demonstrating its capability on large graphs where several methods fail to complete within resource limits.

Label Cost Reduction. Beyond final accuracy, HIAL substantially reduces annotation costs as shown in Figure 4. To reach 55% accuracy on Citeseer, HIAL requires only 48 labeled nodes, while RIM requires 108 nodes. This represents a label budget reduction of over 50%, suggesting that the dual-perspective surrogate identifies informative instances earlier than competitors. Statistical significance is confirmed via paired t -tests ($p < 0.05$).

Table 3. Ablation study quantifying the contribution of each component with label budget $B = 20C$. Red numbers indicate the performance decrease (gap) compared to the full model.

Method	Cora	Citeseer	DBLP	PubMed
HIAL	72.97	61.55	78.41	76.14
w/o <i>MFS</i>	70.23 -2.74	55.78 -5.77	73.61 -4.80	72.45 -3.69
w/o <i>ESR</i>	70.79 -2.18	60.22 -1.33	75.66 -2.75	73.80 -2.34
w/o <i>HOIP</i>	68.84 -4.13	56.47 -5.08	66.75 -11.65	70.52 -5.62

4.3. Model Interpretability (Q2)

To quantify the contribution of each component in HIAL, we conducted an ablation study by isolating the HOI Propagation (HOIP), Feature Sensitivity (MFS), and Structural Reachability (ESR). The results are summarized in Table 3.

Table 4. Diagnostic ablation of HOIP weighting mechanisms while keeping the residual connection unchanged.

Budget	Propagation Variant	Cora	Citeseer	PubMed
5C	CE + residual	56.26	34.21	59.87
	CE + inverse cardinality	58.40	40.70	62.40
	CE + semantic weight	61.07	44.60	61.53
	HOIP	62.27	46.90	66.24
10C	CE + residual	66.80	53.26	69.48
	CE + inverse cardinality	66.91	55.10	71.55
	CE + semantic weight	68.20	55.20	70.60
	HOIP	69.03	56.80	72.99

Excluding MFS. Relying solely on structural reach (ESR) without feature guidance led to a notable decline, particularly on Citeseer (-5.77%) and PubMed (-3.69%). This confirms that structural centrality cannot fully substitute for feature-based information gain. MFS is essential for capturing diffusion-based feature-space coverage, encouraging selected nodes to span diverse regions of the smoothed feature manifold.

Excluding ESR. The exclusion of ESR caused consistent performance drops (e.g., -2.75% on DBLP). While MFS captures feature sensitivity, it depends on propagation quality which may suffer from oversmoothing. ESR serves as a topological reachability component. By modeling collective activation induced by hyperedge co-membership, it encourages locally reachable seeds even when feature-space coverage alone is insufficient, preventing selection from relying solely on global feature similarity.

Disabling Propagation. Removing the HOIP module resulted in the most severe performance degradation across the majority of datasets. Specifically, on DBLP, the accuracy dropped by 11.65%, the largest gap observed. This indicates that raw features alone are insufficient for identifying influential nodes. The HOIP mechanism is foundational, as it smooths features based on interaction intensity to reveal high-order context required for subsequent influence estimation. Its absence leads to an influence assessment based on less informative representation, misjudging the influence potential that arises from network interactions.

Diagnostic HOIP Ablation. To isolate the contribution of HOIP, we further compare against weighted CE variants while keeping the residual connection unchanged. As shown in Table 4, inverse-cardinality correction mainly mitigates structural inflation from large hyperedges, whereas semantic weighting suppresses feature-inconsistent hyperedges. Their combination consistently performs best, supporting that the two corrections are complementary.

4.4. Efficiency and Scalability (Q3)

Another advantage of HIAL is its high efficiency in node selection. As a training-free acquisition method, HIAL avoids the iterative retraining required by model-dependent

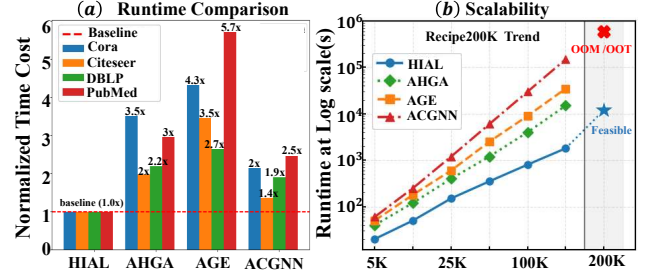


Figure 5. (a) **Active-selection runtime comparison** showing HIAL achieves significant speedup (up to $5.7\times$) against learning-based methods. (b) **Scalability analysis.** We deployed methods on sampled Recipe200K, showing a time-cost trend. The right part is stress testing on full dataset.

approaches during node selection. We evaluate computational efficiency from three perspectives: runtime, trend scalability, and stress testing. The active-selection runtime comparison is presented in Figure 5a.

Efficiency. As shown in Figure 5a, HIAL achieves up to $5.7\times$ speedup compared to learning-based methods (e.g., AHGA, ACGNN). This is attributed to the training-free acquisition design, which eliminates the retraining loops and gradient back-propagation required by model-dependent competitors during selection.

Scalability. Figure 5b highlights the critical advantage of our Lazy Greedy optimization. On the Recipe200K (240k nodes) dataset, HIAL demonstrates superior scalability. In contrast, baselines such as AHGA, RIM, and AGE fail due to OOT or OOM, as shown in Table 2. This confirms that HIAL effectively reduces the repeated training cost typically incurred by model-dependent acquisition on large hypergraphs, making it suitable for real-world deployment.

Stress Testing. To evaluate scalability under extreme conditions, we deployed the model on the full Recipe200K dataset (240K nodes). HIAL completes the process in approximately 2.8 hours (10^4 seconds). In contrast, baselines like AHGA suffer from OOM errors, while AGE and ACGNN may take over 300 hours under the scaling trend.

4.5. Generalization (Q4)

Table 5 evaluates HIAL-style selection when the influence propagation operator is instantiated with different HNN propagation schemes. Details about HNNs' propagation scheme can be found in the Appendix D. We observe two key phenomena:

Superiority of HOIP: HIAL consistently outperforms HIAL-HGNN (using uniform CE). This performance gap empirically validates our analysis in Section 3.2: uniform CE-style propagation is more prone to *ranking collapse*, leading to sub-optimal node selection. In contrast, HOIP retains useful context through cardinality- and consistency-aware weighting, identifying more informative nodes.

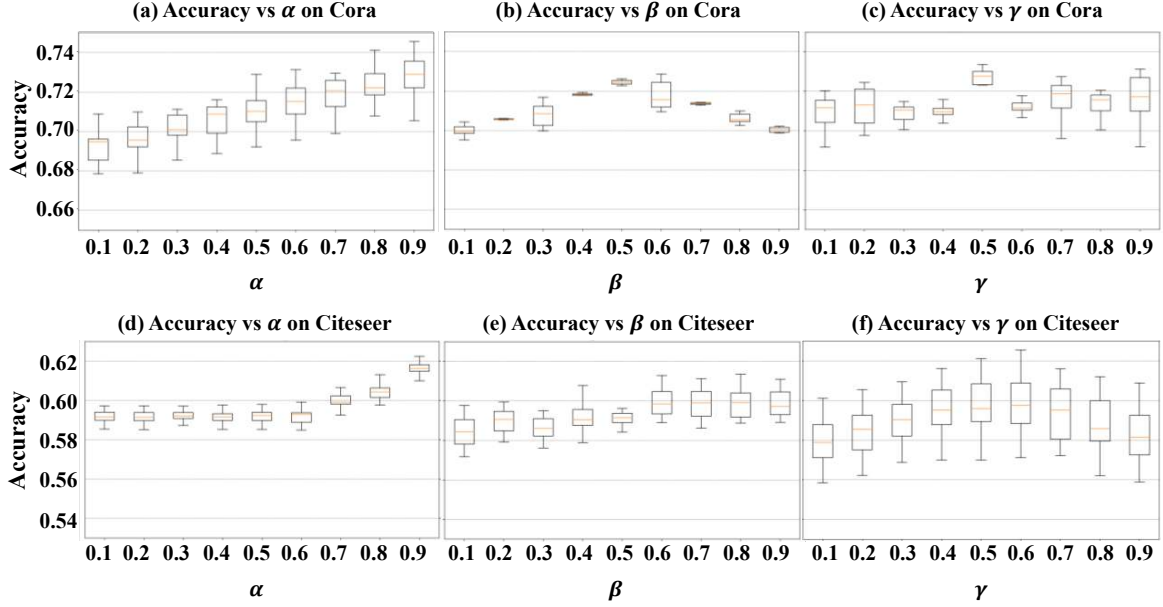


Figure 6. Hyperparameter sensitivity analysis of α , β , and γ on Cora and Citeseer datasets. The box plots illustrate the test accuracy distribution. Note the high stability of β and the contrasting trends of γ , which validate the adaptability of HIAL to different graphs.

Robustness of HIAL Framework: Notably, even HIAL-HGNN significantly outperforms standard baselines. This indicates that our influence maximization framework is robust: with standard clique expansion propagation (HGNN), the global structural perspective provided by the HIAL strategy is more effective than local heuristic-based methods.

We also compare against the strongest influence-based baseline under the same UniGCN backbone. On Cora, HIAL-UniGCN achieves 72.71%, outperforming GRAIN-UniGCN at 70.52%, suggesting that the gain is not solely due to the downstream backbone.

4.6. Robustness (Q5)

Figure 6 illustrates the robustness of HIAL with respect to α , β , and γ . Appendix F further reports sensitivity to the activation threshold θ and coverage radius r .

Parameter α . We observe a positive correlation between α and accuracy on both datasets. This trend indicates that a higher propagation strength effectively aggregates high-order neighborhood information.

Parameter β . On Cora, the performance shows an optimal value around $\beta = 0.5$, suggesting that a moderate ESR transmission probability balances reachability and diffusion decay.

Parameter γ . Crucially, accuracy on Cora is maximized at $\gamma = 0.5$, while Citeseer also maintains optimal performance around the median range. This trend empirically supports the dual-perspective design, supporting that integrating both

Table 5. Generalization analysis of HIAL across different HNN backbones. Underlined values indicate performance that exceeds the second-best baseline reported in Table 2. HIAL-X denotes using HIAL selection with X’s propagation matrix.

Method	Cora	Citeseer	DBLP	PubMed
HIAL-HGNN	70.95	<u>60.20</u>	<u>75.75</u>	<u>75.69</u>
HIAL-UniGCN	<u>72.71</u>	<u>60.13</u>	<u>75.19</u>	<u>75.31</u>
HIAL-HNHN	71.40	60.26	<u>75.42</u>	74.68
HIAL	72.97	61.55	78.41	76.14

structural reachability and feature representativeness yields superior performance compared to relying on a single view ($\gamma \rightarrow 0$ or $\gamma \rightarrow 1$).

5. Conclusion

In this paper, we presented HIAL, a training-free hypergraph active learning framework designed for homophilic hypergraphs. HIAL mitigates the ranking collapse caused by uniformly weighted clique expansion through a high-order-aware weighted pairwise projection and linear diffusion. By combining feature-space coverage with structural reachability, HIAL defines a monotone submodular surrogate objective that can be optimized efficiently via Lazy Greedy selection with a $(1 - 1/e)$ approximation guarantee for the surrogate objective. Empirically, HIAL achieves strong label efficiency, accuracy, and scalability across eight benchmarks. Future work will explore adaptive interaction weighting for heterophilic and dynamic hypergraphs.

Impact Statement

This paper presents work whose goal is to advance efficient active learning on hypergraph-structured data. By reducing annotation requirements, HIAL may lower labeling costs in domains where hypergraphs are homophilic, such as citation, co-authorship, and recommendation networks. A potential limitation is that the method may be less suitable for strongly heterophilic hypergraphs, where feature-inconsistent hyperedges can carry important signals. This limitation is discussed in Appendix H.

Acknowledgements

The authors thank the anonymous reviewers and area chairs for their constructive feedback. This work was supported in part by the National Natural Science Foundation of China under Grant Nos. U24A20255, U2241211, 62427808, 625B2021, and 62572055, and in part by the Guangdong Provincial Ordinary Universities' Special Innovation Program under Grant No. 2024KTSCX260. Rong-hua Li is the corresponding author of this paper.

References

- Cai, H., Zheng, V. W., and Chang, K. C.-C. Active learning for graph embedding. *arXiv preprint arXiv:1705.05085*, 2017.
- Chien, E., Pan, C., Peng, J., and Milenkovic, O. You are AllSet: A multiset learning framework for hypergraph neural networks. In *International Conference on Learning Representations*, 2022.
- Chien, I. E., Zhou, H., and Li, P. HS²: Active learning over hypergraphs with pointwise and pairwise queries. In *Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics*, pp. 2466–2475. PMLR, 2019.
- Ding, K., Wang, J., Li, J., Li, D., and Liu, H. Be more with less: Hypergraph attention networks for inductive text classification. In *Proceedings of the 2020 conference on empirical methods in natural language processing (EMNLP)*, pp. 4927–4936, 2020.
- Dong, Y., Sawin, W., and Bengio, Y. HNNH: Hypergraph networks with hyperedge neurons. *arXiv preprint arXiv:2006.12278*, 2020.
- Dua, D. and Graff, C. Uci machine learning repository. <http://archive.ics.uci.edu/ml>, 2017.
- Feng, Y., You, H., Zhang, Z., Ji, R., and Gao, Y. Hypergraph neural networks. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, pp. 3558–3565, 2019.
- Guillory, A. and Bilmes, J. Active semi-supervised learning using submodular functions. In *Proceedings of the Twenty-Seventh Conference on Uncertainty in Artificial Intelligence*, pp. 274–282, 2011.
- He, X., Deng, K., Wang, X., Li, Y., Zhang, Y., and Wang, M. LightGCN: Simplifying and powering graph convolution network for recommendation. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 639–648, 2020.
- Huang, J. and Yang, J. UniGNN: a unified framework for graph and hypergraph neural networks. In *Proceedings of the Thirtieth International Joint Conference on Artificial Intelligence*, pp. 2563–2569. International Joint Conferences on Artificial Intelligence Organization, 2021.
- Kempe, D., Kleinberg, J., and Tardos, É. Maximizing the spread of influence through a social network. In *Proceedings of the ninth ACM SIGKDD international conference on Knowledge discovery and data mining*, pp. 137–146, 2003.
- Kim, S., Lee, S. Y., Gao, Y., Antelmi, A., Polato, M., and Shin, K. A survey on hypergraph neural networks: An in-depth and step-by-step guide. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pp. 6534–6544, 2024.
- Lang, K. Newsweeder: Learning to filter netnews. In *Machine learning proceedings 1995*, pp. 331–339. Elsevier, 1995.
- Li, S., Li, Y., Ni, J., and McAuley, J. SHARE: a system for hierarchical assistive recipe editing. In *Proceedings of the 2022 conference on empirical methods in natural language processing*, pp. 11077–11090, 2022.
- Liao, X., Tu, B., Li, J., and Plaza, A. Class-wise graph embedding-based active learning for hyperspectral image classification. *IEEE Transactions on Geoscience and Remote Sensing*, 61:1–13, 2023.
- Ma, J., Ma, Z., Chai, J., and Mei, Q. Partition-based active learning for graph neural networks. *Transactions on Machine Learning Research*, 2023. ISSN 2835-8856. Survey Certification.
- Minoux, M. Accelerated greedy algorithms for maximizing submodular set functions. In *Optimization Techniques: Proceedings of the 8th IFIP Conference on Optimization Techniques Würzburg, September 5–9, 1977*, pp. 234–243. Springer, 2005.
- Nemhauser, G. L., Wolsey, L. A., and Fisher, M. L. An analysis of approximations for maximizing submodular set functions—i. *Mathematical Programming*, 14(1): 265–294, 1978.

- Ni, X., Mei, G., Zhang, S.-S., Chen, Y., Xu, X., Liu, C., and Zhan, X.-X. Structural-aware key node identification in hypergraphs via representation learning and fine-tuning. *Engineering Applications of Artificial Intelligence*, 169: 114108, 2026.
- Rossi, E., Frasca, F., Chamberlain, B., Eynard, D., Bronstein, M., and Monti, F. SIGN: Scalable inception graph neural networks. *arXiv preprint arXiv:2004.11198*, 2020.
- Song, Z., Zhang, Y., and King, I. No change, no gain: Empowering graph neural networks with expected model change maximization for active learning. In *Advances in Neural Information Processing Systems*, volume 36, pp. 47511–47526, 2023.
- Sun, Y., Qin, A., Bandoh, Y., Gao, C., and Hiwasaki, Y. Active learning for hyperspectral image classification via hypergraph neural network. In *2022 IEEE International Conference on Image Processing (ICIP)*, pp. 2576–2580. IEEE, 2022.
- Tang, B., Liu, Z., Jiang, K., Chen, S., and Dong, X. Training-free message passing for learning on hypergraphs. In *International Conference on Learning Representations*, volume 2025, pp. 69491–69519, 2025.
- Wang, H. and Leskovec, J. Unifying graph convolutional neural networks and label propagation. *arXiv preprint arXiv:2002.06755*, 2020.
- Xie, M., Zhan, X.-X., Liu, C., and Zhang, Z.-K. Influence maximization in hypergraphs, 2022. URL <https://arxiv.org/abs/2206.01394>.
- Xu, K., Li, C., Tian, Y., Sonobe, T., Kawarabayashi, K.-i., and Jegelka, S. Representation learning on graphs with jumping knowledge networks. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pp. 5453–5462. PMLR, 2018.
- Yadati, N., Nimishakavi, M., Yadav, P., Nitin, V., Louis, A., and Talukdar, P. HyperGCN: A new method for training graph convolutional networks on hypergraphs. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- Zhang, W., Wang, Y., You, Z., Cao, M., Huang, P., Shan, J., Yang, Z., and Cui, B. RIM: Reliable influence-based active learning on graphs. In *Advances in Neural Information Processing Systems*, volume 34, pp. 27978–27990, 2021a.
- Zhang, W., Yang, Z., Wang, Y., Shen, Y., Li, Y., Wang, L., and Cui, B. GRAIN: Improving data efficiency of graph neural networks via diversified influence maximization. *Proceedings of the VLDB Endowment*, 2021b.

A. Proof

Lemma A.1. *The activation function $\sigma(\cdot)$ satisfies the union property: for any two sets $A, B \subseteq \mathcal{V}$, $\sigma(A \cup B) = \sigma(A) \cup \sigma(B)$.*

Proof. Recall the definition

$$\sigma(S) = \{j \in \mathcal{V} \mid \max_{u \in S} I_j(u, k) > \theta\}.$$

For any node j , the condition $j \in \sigma(A \cup B)$ is equivalent to

$$\max_{u \in A \cup B} I_j(u, k) > \theta.$$

By the property of the maximum operator,

$$\max_{u \in A \cup B} (\cdot) = \max \left(\max_{u \in A} (\cdot), \max_{u \in B} (\cdot) \right).$$

Thus, the inequality holds if and only if

$$\left(\max_{u \in A} I_j(u, k) > \theta \right) \vee \left(\max_{u \in B} I_j(u, k) > \theta \right).$$

This is logically equivalent to $j \in \sigma(A) \vee j \in \sigma(B)$, i.e., $j \in \sigma(A) \cup \sigma(B)$. Therefore, $\sigma(A \cup B) = \sigma(A) \cup \sigma(B)$. $\square$

Theorem A.2. *The count of activated nodes $f(S) = |\sigma(S)|$ is monotone non-decreasing and submodular.*

Proof. **Monotonicity:** Let $S \subseteq T$. We can express T as $S \cup (T \setminus S)$. By Lemma A.1,

$$\sigma(T) = \sigma(S) \cup \sigma(T \setminus S).$$

Since $A \subseteq A \cup B$ for any sets, it follows that $\sigma(S) \subseteq \sigma(T)$, and hence

$$|\sigma(S)| \leq |\sigma(T)|.$$

Submodularity: Consider $S \subseteq T$ and a node $v \notin T$. Define the marginal gain

$$\Delta_v(S) = |\sigma(S \cup \{v\})| - |\sigma(S)|.$$

Using Lemma A.1, $\sigma(S \cup \{v\}) = \sigma(S) \cup \sigma(\{v\})$. Then

$$\Delta_v(S) = |\sigma(S) \cup \sigma(\{v\})| - |\sigma(S)| = |\sigma(\{v\})| - |\sigma(S) \cap \sigma(\{v\})| = |\sigma(\{v\}) \setminus \sigma(S)|.$$

Since $S \subseteq T$ implies $\sigma(S) \subseteq \sigma(T)$ (by monotonicity),

$$\sigma(\{v\}) \setminus \sigma(T) \subseteq \sigma(\{v\}) \setminus \sigma(S),$$

and taking cardinalities yields $\Delta_v(T) \leq \Delta_v(S)$. Thus, $f(S) = |\sigma(S)|$ is submodular. $\square$

Theorem A.3. *The Magnitude of Feature Sensitivity, $MFS(S)$, is monotone non-decreasing and submodular.*

Proof. Define the set cover function

$$g(X) = \left| \bigcup_{u \in X} G_u \right|,$$

where $X \subseteq \mathcal{V}$ is a node set and G_u denotes the feature-space ball covered by node u . Then the objective can be written as

$$MFS(S) = g(\sigma(S)).$$

Monotonicity: Let $S \subseteq T$. From Theorem A.2, $\sigma(S) \subseteq \sigma(T)$. Since union is monotone, i.e.,

$$A \subseteq B \implies \bigcup_{x \in A} G_x \subseteq \bigcup_{x \in B} G_x,$$

we have $g(\sigma(S)) \leq g(\sigma(T))$, hence $MFS(S)$ is monotone.

Submodularity: It is well-known that the set cover function $g(\cdot)$ is submodular. Consider $S \subseteq T$ and a node $v \notin T$. The marginal gain is

$$\Delta_{MFS}(S, v) = g(\sigma(S \cup \{v\})) - g(\sigma(S)).$$

By Lemma A.1,

$$\sigma(S \cup \{v\}) = \sigma(S) \cup \sigma(\{v\}).$$

Let $X = \sigma(S)$, $Y = \sigma(T)$, and $Z = \sigma(\{v\})$. Since $S \subseteq T$, we have $X \subseteq Y$. By submodularity of $g(\cdot)$, for any $X \subseteq Y$ and any Z ,

$$g(X \cup Z) - g(X) \geq g(Y \cup Z) - g(Y).$$

Applying this inequality yields $\Delta_{MFS}(S, v) \geq \Delta_{MFS}(T, v)$, proving submodularity. $\square$

Theorem A.4. *The Expected Structure Reachability, $ESR(S)$, is monotone non-decreasing and submodular.*

Proof. We first clarify the neighborhood notation to avoid double counting. Let $\mathcal{N}(v)$ denote the neighbor set of v (excluding itself), and define

$$\mathcal{N}(S) = \left(\bigcup_{u \in S} \mathcal{N}(u) \right) \setminus S.$$

Recall

$$ESR(S) = |S| + \sum_{v \in \mathcal{N}(S)} P(v, S), \quad P(v, S) = 1 - \prod_{u \in S \cap \mathcal{N}(v)} (1 - p_{uv}).$$

Assume $p_{uv} \in [0, 1]$ for all (u, v) so that each factor $(1 - p_{uv}) \in [0, 1]$.

Monotonicity: Consider adding a node $w \notin S$. For convenience, define $P(v, S) = 0$ if $S \cap \mathcal{N}(v) = \emptyset$. The marginal gain can be written as

$$\Delta(S, w) = 1 - P(w, S) + \sum_{v \in \mathcal{N}(w) \setminus S} (P(v, S \cup \{w\}) - P(v, S)).$$

The first term is non-negative since $P(w, S) \leq 1$. For each $v \in \mathcal{N}(w) \setminus S$, define

$$Q(v, S) = \prod_{u \in S \cap \mathcal{N}(v)} (1 - p_{uv}).$$

Then

$$P(v, S \cup \{w\}) - P(v, S) = Q(v, S)p_{wv} \geq 0.$$

Thus $\Delta(S, w) \geq 0$, proving monotonicity.

Submodularity: Consider $S \subseteq T$ and $w \notin T$. Since $P(w, S) \leq P(w, T)$, we have

$$1 - P(w, S) \geq 1 - P(w, T).$$

Moreover, for each v , $Q(v, S) \geq Q(v, T)$ because T contains all factors in S and possibly additional factors in $[0, 1]$. Therefore,

$$Q(v, S)p_{wv} \geq Q(v, T)p_{wv}.$$

Together with $\mathcal{N}(w) \setminus T \subseteq \mathcal{N}(w) \setminus S$, the marginal gain under S is no smaller than that under T . Hence $\Delta(S, w) \geq \Delta(T, w)$, proving submodularity. $\square$

B. Dataset

We evaluate the performance of our proposed method on eight widely used benchmark datasets, covering citation networks, co-authorship, social review, biological attributes, text classification, and culinary networks. The detailed statistics are summarized in Table 6.

- **Cora, Citeseer, and PubMed** (Huang & Yang, 2021) are three standard citation network datasets. Nodes represent scientific papers, and node features are bag-of-words vectors. Hyperedges are constructed based on citation relationships, where a hyperedge connects a paper and all the papers it cites (or is cited by), capturing the local citation community.
- **DBLP** (Yadati et al., 2019) is a co-authorship network. Nodes represent papers, and hyperedges represent authors. Specifically, a hyperedge connects all papers co-authored by a specific researcher. The node features are bag-of-words from paper abstracts.
- **Yelp3K** (Chien et al., 2022) is a social comment network. Nodes denote businesses (e.g., restaurants), and features include location, rating, and check-in information. Hyperedges are constructed based on users: a set of businesses reviewed by the same user forms a hyperedge.
- **Mushroom** ((Dua & Graff, 2017)) is a biological dataset from the UCI Repository. Nodes represent mushroom instances, and features are categorical attributes. Hyperedges are constructed by attribute matching: instances sharing the same categorical attribute value form a hyperedge.
- **News20** (Lang, 1995) is a classic text classification dataset. Nodes represent news documents, and features are TF-IDF vectors. Hyperedges link documents that share significant keywords or belong to the same latent topic.
- **Recipe200K** (Li et al., 2022) is a large-scale culinary dataset. Nodes represent recipes, and features are generated from sentence embeddings. Hyperedges are defined by ingredients: recipes sharing a specific ingredient are connected.

Dataset Properties and Homophily. The selected datasets encompass a diverse range of domains including citation networks, co-authorship, social review, biological attributes, text classification, and culinary networks. Crucially, these benchmarks are commonly used in hypergraph learning (Chien et al., 2022; Ding et al., 2020) and generally exhibit meaningful feature or label smoothness within hyperedges, where hyperedge co-membership often reflects shared topical, semantic, behavioral, or attribute-level regularities. This characteristic aligns with our proposed HOI-aware propagation mechanism, which leverages feature consistency to reinforce intra-class connections.

Table 6. Detailed statistics of the eight benchmark datasets. $|\mathcal{V}|$ and $|\mathcal{E}|$ denote the number of nodes and hyperedges, respectively. d represents the dimension of node features, and C is the number of classes. The Split column indicates the number of samples used for Train/Validation/Test.

Dataset	$ \mathcal{V} $	$ \mathcal{E} $	d	C	Split (Train/Val/Test)	Domain
Cora	2,708	1,579	1,433	7	1408/500/800	Co-citation
Citeseer	3,327	1,079	3,703	6	2027/500/800	Co-citation
PubMed	19,717	7,963	500	3	18417/500/800	Co-citation
DBLP	4,057	14,328	334	4	2757/500/800	Co-authorship
Yelp3K	3,855	24,137	1,862	6	2555/500/800	Social Review
News20	16,342	100	100	20	15042/500/800	Text Classification
Mushroom	8,124	298	22	2	6824/500/800	Biological Attributes
Recipe200K	240,094	18,129	3,200	8	175K/25K/40K	Culinary Network

C. Baselines

We compare HIAL with three categories of baselines: Heuristic methods, Graph Active Learning (GAL) methods, and Hypergraph Active Learning (HAL) methods.

Heuristics

- **Random** Select nodes randomly.
- **Degree-N** Select nodes with highest node degree.

- **Degree-HE** Select nodes with highest hyperedge degree.

GAL Methods. For these methods, we transform the hypergraph into a graph via clique expansion.

- **AGE** (Cai et al., 2017): An active graph embedding framework that linearly combines three metrics: centrality, informativeness (entropy), and representativeness. It employs time-sensitive parameters to dynamically adjust the weight of these metrics, shifting focus from graph structure to embedding uncertainty during training.
- **GRAIN** (Zhang et al., 2021b): A framework that bridges active learning with social influence maximization. It treats GNN feature propagation as an influence process and selects nodes using a Diversified Influence Maximization (DIM) criterion to maximize the number of influenced unlabeled nodes while ensuring diversity.
- **RIM** (Zhang et al., 2021a): An extension of influence-based methods designed for noisy label settings. It proposes a Reliable Influence Maximization criterion that simultaneously considers the quantity (reach) and quality (reliability) of influence to mitigate the propagation of label noise.
- **GraphPart** (Ma et al., 2023): A partition-based active learning approach. It first decomposes the graph into disjoint communities and then selects representative nodes within each partition, ensuring structural coverage without introducing additional hyperparameters.
- **DOCTOR** (Song et al., 2023): A method based on the Expected Model Change Maximization principle. It derives a closed-form acquisition function using a lightweight Bayesian interpretation of GCNs, selecting nodes that maximize the expected change in model predictions without retraining.

HAL Methods.

- **ACGNN** (Sun et al., 2022): A method for hyperspectral image classification that constructs a hypergraph to capture non-pairwise pixel relationships. It iteratively selects pixels based on local hypergraph structural properties to fine-tune a semi-supervised hypergraph neural network.
- **CGE-AL** (Liao et al., 2023): A class-wise graph embedding framework for hyperspectral images. It utilizes a Class-wise GCN to measure uncertainty between labeled and unlabeled nodes, querying samples based on a class-minimum uncertainty strategy.
- **AHGA** (Ni et al., 2026): A framework for key node identification in hypergraphs. It integrates an autoencoder for structural feature extraction and employs an active learning-based fine-tuning process to bridge the gap between synthetic pre-training data and real-world hypergraph topologies.

Note on Computational Efficiency (OOM/OOT): In our experiments, OOM denotes exceeding 24GB GPU memory, and OOT denotes exceeding a 24-hour wall-clock limit. As observed in Table 2 of the main text, many baselines (e.g., AHGA, RIM, AGE) fail on large-scale datasets like Recipe200K due to Out-of-Memory (OOM) or Out-of-Time (OOT) errors. This is primarily because model-dependent methods (like ACGNN) require *retraining the entire HNN model* from scratch after every batch of node selection to update uncertainty scores. For a graph with hundreds of thousands of nodes, this iterative retraining is computationally prohibitive. In contrast, HIAL is a training-free framework that computes influence scores through sparse propagation and structural reachability, employing a Lazy Greedy algorithm that avoids redundant computations. This design difference allows HIAL to rely on sparse hypergraph operations, whereas retraining-based competitors incur substantially higher computational costs or excessive memory usage.

D. HNN Models

To verify the generalization capability of HIAL (Q4 in Experiments), we integrated our selection strategy with different downstream HNN backbones. Since HIAL is decoupled from downstream training, its acquisition module can be instantiated with the propagation operator of different HNN backbones. The backbones used are:

- **HGNN** (Feng et al., 2019): Based on hypergraph spectral theory, it approximates the convolution on the hypergraph Laplacian. It employs symmetric normalization ($\mathbf{D}^{-1/2}$) to perform a node-edge-node transform, capturing high-order correlations through node-hyperedge-node propagation.

- **UniGCN** (Huang & Yang, 2021): Derived from a unified framework bridging GNNs and HNNs, it generalizes the isotropic GCN propagation to hypergraphs. It introduces specific normalization terms involving $\bar{\mathbf{D}}_e$ to balance the impact of varying hyperedge sizes during message passing.
- **HNHN** (Dong et al., 2020): It treats hyperedges as explicit computational neurons rather than mere transmission channels. Unlike spectral methods, HNHN utilizes random walk normalization ($\mathbf{D}^{-1}$), interpreting the propagation as a probabilistic transition between nodes and hyperedges.

The specific propagation formulas for these backbones are listed in Table 7. Note that for HIAL-X variants, we use the corresponding propagation matrix to compute the Feature Sensitivity (MFS), ensuring the selection metric aligns with the downstream model’s topology.

Table 7. The decoupled propagation mechanisms of various HNN backbones used in generalization experiments. $\bar{\mathbf{D}}_e$ denotes the average degree of nodes inside a hyperedge.

HNN Model	Propagation Mechanism Formula
HGNN	$\mathbf{X}^{(k)} = \mathbf{D}_v^{-1/2} \mathbf{H} \mathbf{W} \mathbf{D}_e^{-1} \mathbf{H}^\top \mathbf{D}_v^{-1/2} \mathbf{X}^{(k-1)}$
UniGCN	$\mathbf{X}^{(k)} = \mathbf{D}_v^{-1/2} \mathbf{H} \mathbf{W} \bar{\mathbf{D}}_e^{-1/2} \mathbf{H}^\top \mathbf{D}_e^{-1} \mathbf{X}^{(k-1)}$
HNHN	$\mathbf{X}^{(k)} = \mathbf{D}_v^{-1} \mathbf{H} \mathbf{W} \mathbf{D}_e^{-1} \mathbf{H}^\top \mathbf{X}^{(k-1)}$

E. Experiment Implementations

- **Model Architecture:** To ensure a fair comparison, we use a standard 2-layer decoupled HGNN with a hidden dimension of 64 as the downstream classifier for all selection strategies. The dropout rate is set to 0.6, the learning rate to 0.01, and weight decay to 5×10^{-4} .
- **HIAL Hyperparameters:**
 - The activation threshold is set to $\theta = 0.25$.
 - The coverage radius r is selected according to the scale of the normalized cosine distance; unless otherwise stated, we use $r = 0.10$.
 - The propagation coefficient α in HOIP is set to 0.9.
 - The ESR transmission probability β is set to 0.5.
 - The balance factor γ in unified function $F(S)$ is set to 0.5, reflecting equal weighting between structural reachability and feature-space coverage.
 - The propagation depth is set to $k = 1$.
- **Baselines Setup:** For open-source methods (AGE, GRAIN, RIM), we utilized their official implementations. For ACGNN and CGE-AL, we followed the settings reported in their original papers. To ensure reliable uncertainty estimation for model-based methods, we allocated 200 training epochs in each active learning iteration.
- **Hardware:** All experiments were conducted on a server running Ubuntu 20.04, 16 Intel(R) Xeon(R) Gold 6430 CPU cores and an NVIDIA RTX 4090 GPU. The code is written in Python 3.8 with PyTorch 1.11.0 and CUDA 11.3.

Practical parameter selection. In addition to the fixed defaults used in our experiments, HIAL admits simple data-driven settings. For the coverage radius r , one can set it to the average cosine distance from each node to its q -th nearest neighbor in the propagated feature space, which adapts the coverage scale to the intrinsic feature density. For the activation threshold θ , a robust choice is the 75th percentile of non-zero normalized influence scores $I_j(i, k)$, which keeps only relatively strong diffusion signals while avoiding dataset-specific absolute thresholds.

F. Hyperparameter Sensitivity Analysis

In the main text, we adopted fixed values for hyperparameters (e.g., $\alpha = 0.9, \gamma = 0.5, \beta = 0.5$) to ensure the usability and simplicity of HIAL without requiring extensive tuning. To validate these choices, we evaluated the performance variance by sweeping these parameters across a broad range.

Balancing Parameter Stability (α, β, γ): We further analyze the robustness of HIAL regarding its key components and hyperparameters.

1. Impact of Propagation Weight (α): HIAL exhibits a **positive correlation** with α on both datasets (Figs 6a and d).

- On **Cora**, accuracy steadily climbs from ~ 0.69 at $\alpha = 0.1$ to a peak of ~ 0.73 at $\alpha = 0.9$.
- On **Citeseer**, performance is stable for $\alpha \in [0.1, 0.6]$ but shows a sharp improvement as α approaches 0.9.
- *Interpretation:* This trend supports the importance of the proposed propagation mechanism: assigning a higher weight to the proposed influence propagation mechanism (relative to the base initialization) significantly boosts performance.

2. Impact of ESR Transmission Probability (β): The parameter β shows a moderate optimum:

- On **Cora** (Fig 6b), there is a moderate optimum peak at $\beta = 0.5$, suggesting that a moderate transmission probability balances reachability and diffusion decay.
- On **Citeseer** (Fig 6e), the performance is relatively robust but shows a slight preference for $\beta \geq 0.6$.

3. Impact of Component Balance (γ): The balance parameter γ confirms the necessity of integrating both perspectives:

- On **Cora** (Fig 6c), the accuracy peaks noticeably at $\gamma = 0.5$, indicating that structural reachability and feature-space coverage are complementary and equally important.
- On **Citeseer** (Fig 6f), while the variance is higher, the median accuracy remains stable across the range, demonstrating HIAL’s adaptability to different topological densities.

Sensitivity to Threshold θ and Radius r : We also evaluated the sensitivity of the activation threshold θ and feature radius r . We found that performance is stable for $\theta \in [0.15, 0.35]$ and $r \in [0.05, 0.20]$. Extreme values (e.g., $\theta > 0.5$) lead to sparse activation sets, reducing the effectiveness of MFS, while $r \rightarrow \infty$ trivializes the diversity constraint. The default values ($\theta = 0.25, r = 0.1$) provide a balanced trade-off across all datasets.

G. Time Complexity Analysis

Notation. Let $n = |\mathcal{V}|$, $m = |\mathcal{E}|$, and $N_{nz} = \sum_{e \in \mathcal{E}} |e|$ be the number of nonzero entries in the incidence matrix $\mathbf{H}$. Let F denote the feature dimension, k the propagation depth, and B the labeling budget. By default, HIAL uses $k = 1$.

HOIP weight construction. HIAL first computes the hyperedge-level semantic dispersion σ_e^2 for each hyperedge. This requires one traversal over incident node features and costs $O(N_{nz}F)$ time. The structural factor $(d(e) - 1)^{-1}$ and semantic factor $\exp(-\sigma_e^2)$ are stored as hyperedge-level weights, requiring $O(m)$ additional memory.

Although HOIP is written in node–node form in Eq. (5), the induced weights are produced from hyperedge-level scalars. Therefore, propagation can be implemented by sparse node–hyperedge–node operations without dense clique materialization. If one explicitly caches the sparse induced node–node operator, let M_c denote the number of nonzero co-occurring node pairs. Then the cached operator requires $O(M_c)$ memory, and explicit pair enumeration costs up to $O(\sum_{e \in \mathcal{E}} |e|^2)$ in the worst case. This is optional; HIAL does not require dense $O(n^2)$ storage.

HOI-aware propagation. Each propagation step applies $\mathbf{X}^{(t+1)} = \alpha \tilde{\mathbf{L}} \mathbf{X}^{(t)} + (1 - \alpha) \mathbf{X}^{(0)}$. Using incidence-based sparse multiplication, one step costs $O(N_{nz}F)$, so k steps cost $O(kN_{nz}F)$. If a cached sparse node–node operator is used, the corresponding cost is $O(kM_cF)$. In either implementation, the dense $O(n^2F)$ clique-expanded propagation is avoided.

Sensitivity evaluation. Because HOIP is linear, the propagated features satisfy $\mathbf{X}^{(k)} = \mathbf{A}_k \mathbf{X}^{(0)}$, where $\mathbf{A}_k$ is the effective diffusion operator in Eq. (7). For node features of dimension F , the block Jacobian from node i to node j is

$$\frac{\partial \mathbf{X}_j^{(k)}}{\partial \mathbf{X}_i^{(0)}} = [\mathbf{A}_k]_{ji} \mathbf{I}_F.$$

Thus the sensitivity score in Eq. (8) can be obtained directly from the diffusion operator, without back-propagation or dense Jacobian construction.

For a candidate node v , the influence column is computed by applying the effective diffusion recurrence to a one-hot vector $\mathbf{e}_v$. This costs $O(kN_{nz})$ with incidence-based multiplication, or $O(kM_c)$ with a cached sparse operator. The row-normalization denominator in Eq. (9) is precomputed once by applying the same recurrence to the all-one vector, with the same asymptotic cost.

MFS and ESR evaluation. MFS operates on the propagated features $\mathbf{X}^{(k)}$. Coverage sets G_u are evaluated by cosine similarity in the propagated feature space and do not require additional propagation. Let C_G denote the total cost of constructing or retrieving these coverage sets during Lazy Greedy evaluation. In implementation, coverage sets are computed lazily and reused when possible.

ESR only depends on one-hop co-membership neighborhoods. Let $d_N(v) = |\mathcal{N}(v)|$. The ESR marginal gain for candidate v costs $O(d_N(v))$, and the total ESR cost over all evaluated candidates is $O(\sum_{v \in \mathcal{Q}} d_N(v))$, where $\mathcal{Q}$ is the multiset of candidates re-evaluated by Lazy Greedy.

Lazy Greedy selection. Let Q be the total number of marginal-gain evaluations performed by Lazy Greedy across B rounds. A naive greedy algorithm requires $O(Bn)$ evaluations, while Lazy Greedy substantially reduces this number in practice. Under incidence-based multiplication, the total selection cost is $O(QkN_{nz} + C_G + \sum_{v \in \mathcal{Q}} d_N(v))$. With a cached sparse node–node operator, the first term becomes $O(QkM_c)$.

Overall complexity. Combining all stages, the incidence-based implementation has total time complexity

$$O\left(N_{nz}F + kN_{nz}F + kN_{nz} + QkN_{nz} + C_G + \sum_{v \in \mathcal{Q}} d_N(v)\right).$$

Since $k = 1$ by default, this is dominated by sparse propagation, Lazy Greedy influence evaluation, and feature-space coverage evaluation. The key advantage is that HIAL avoids dense clique-expanded matrices and does not require iterative downstream HNN retraining during acquisition.

The basic memory complexity is $O(N_{nz} + nF + m + n)$, accounting for the sparse incidence structure, node features, propagated features, hyperedge weights, Lazy Greedy queue, and selection indicators. Optional sparse caches for the induced node–node operator or MFS coverage sets require additional storage proportional to their nonzero entries.

H. Limitations and Future Work

While HIAL demonstrates strong performance across standard benchmarks, we recognize a critical limitation regarding its underlying topological assumption.

Dependence on Homophily Assumption: The core design of the HOI-aware propagation mechanism (specifically the semantic gate in Eq. 5) relies on the assumption of homophily. By employing the exponential attenuation term $\exp(-\sigma_e^2)$, our method explicitly penalizes hyperedges with high feature variance, treating them as noise or weak connections.

While this effectively filters structural noise in homophilic datasets (e.g., citation networks), it presents a challenge for heterophilic hypergraphs (e.g., dating or transaction networks), where interactions often occur between dissimilar nodes. In such cases, high intra-hyperedge variance represents critical structural information rather than noise. Consequently, HIAL poses a risk of over-smoothing or severing informative heterophilic connections.

Mitigation and Future Directions: Currently, HIAL partially mitigates this issue through the MFS module, which encourages diversity in the propagated feature space and can partially reduce over-reliance on local structural reachability. However, the structural propagation remains biased towards homophily.

A promising direction for future work is to extend the interaction intensity function to be **adaptive to heterophily**. For instance, replacing the fixed variance penalty with a learnable attention mechanism, a signed interaction function, or a heterophily-aware class-conditional metric could allow the model to distinguish between “noisy variance” and “informative heterophily,” thereby broadening the applicability of HIAL to disassortative mixing patterns.